

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{R_s}{r}} + r^2 d\Omega^2$$

$\uparrow$ $d\theta^2 + \sin^2\theta d\varphi^2$

Reducing a 2 body problem.

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) - V(r)$$

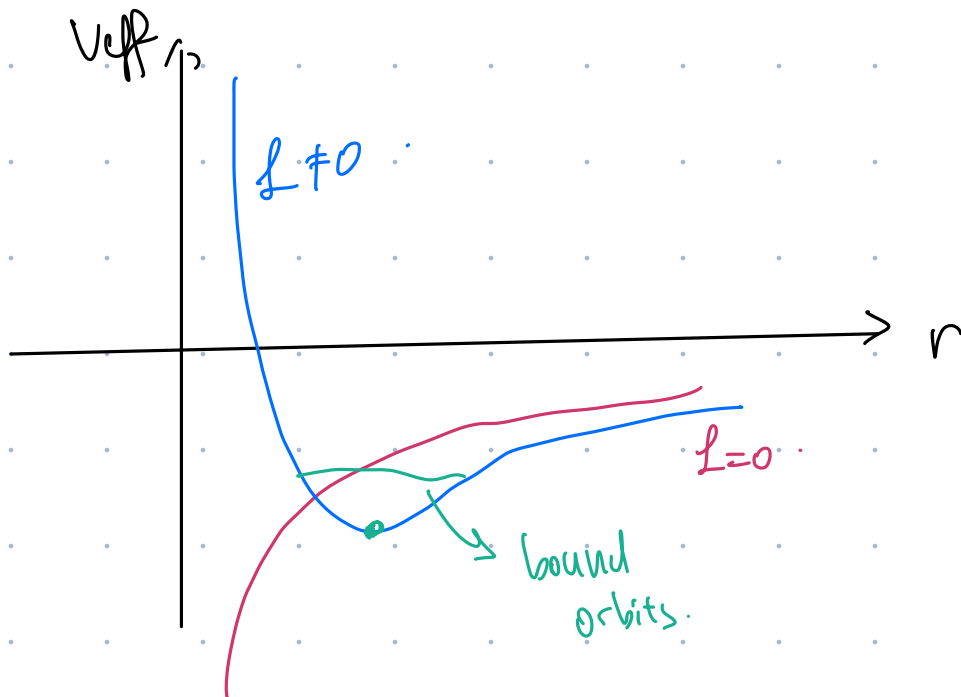
$\nwarrow \frac{G_N M}{r}$

$$L = m r^2 \dot{\varphi}, \quad E = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) + V(r)$$

$$\begin{aligned} \hookrightarrow \dot{\varphi} &= \frac{L}{m r^2} \\ &= \frac{1}{2}m\left(\dot{r}^2 + \cancel{r^2} \frac{L^2}{m^2 \cancel{r^2}}\right) + V(r) \\ &= \frac{1}{2}m\left(\dot{r}^2 + \frac{L^2}{m^2 r^2}\right) + V(r) \\ &= \underbrace{\frac{1}{2}m\dot{r}^2}_{\text{Kinetic energy}} + \underbrace{\left[\frac{1}{2} \frac{L^2}{m r^2} + V(r)\right]}_{\text{Effective Potential } V_{\text{eff}}} \end{aligned}$$

$$R_s = 2 G_N M$$

Set $m=1$: $V_{\text{eff}} = \frac{L^2}{2r^2} - \frac{G_N M}{r} = \frac{1}{2} \left(\frac{L^2}{r^2} - \frac{R_s}{r} \right)$



$$\frac{1}{2} \cdot r^{-2}$$

~~0~~

$$\frac{1}{2} \cdot (-2) r^{-3}$$

$$\frac{dV_{\text{eff}}}{dr} = -\frac{2L^2}{r^3} + \frac{R_s}{r^2}$$

$$-2L^2 + R_s r_{\min} = 0 \Rightarrow r_{\min} = \frac{2R_s}{R_s}$$

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{R_s}{r}} + r^2 d\Omega^2$$

$\nwarrow d\theta^2 + \sin^2\theta d\varphi^2$

↪ conserve energy & angular momentum.

constant along ↙

$$\boxed{\bar{E} = \left(1 - \frac{R_s}{r}\right) \frac{dt}{d\lambda}}$$

$$L = r^2 \frac{d\varphi}{d\lambda}$$

Pick $\theta = \frac{\pi}{2}$: New metric

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{R_s}{r}} + \cancel{r^2 d\theta^2} + r^2 d\varphi^2$$

$$\left(1 - \frac{R_s}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 - \frac{\left(\frac{dr}{d\lambda}\right)^2}{1 - \frac{R_s}{r}} - r^2 \left(\frac{d\varphi}{d\lambda}\right)^2 = \begin{cases} 1, & \lambda = 2, m \neq 0 \\ 0, & \text{when } m = 0. \end{cases}$$

$= \epsilon$

$$\cancel{\left(1 - \frac{R_s}{r}\right)} \frac{E^2}{\left(1 - \frac{R_s}{r}\right)^2} - \frac{\left(\frac{dr}{d\lambda}\right)^2}{\left(1 - \frac{R_s}{r}\right)} - r^2 \frac{L^2}{r^4} = \epsilon$$

$$E^2 - \left(\frac{dr}{d\lambda}\right)^2 - \frac{L^2}{r^2} \left(1 - \frac{R_S}{r}\right) = \epsilon \left(1 - \frac{R_S}{r}\right)$$

$$\left(\frac{dr}{d\lambda}\right)^2 + \left(\frac{L^2}{r^2} + \epsilon\right) \left(1 - \frac{R_S}{r}\right) - E^2 = 0$$

$$\underbrace{\left(\frac{dr}{d\lambda}\right)^2}_{KE} + \underbrace{\left(\frac{L^2}{r^2} + \epsilon\right) \left(1 - \frac{R_S}{r}\right)}_{\text{Effective Potential}} = E^2$$

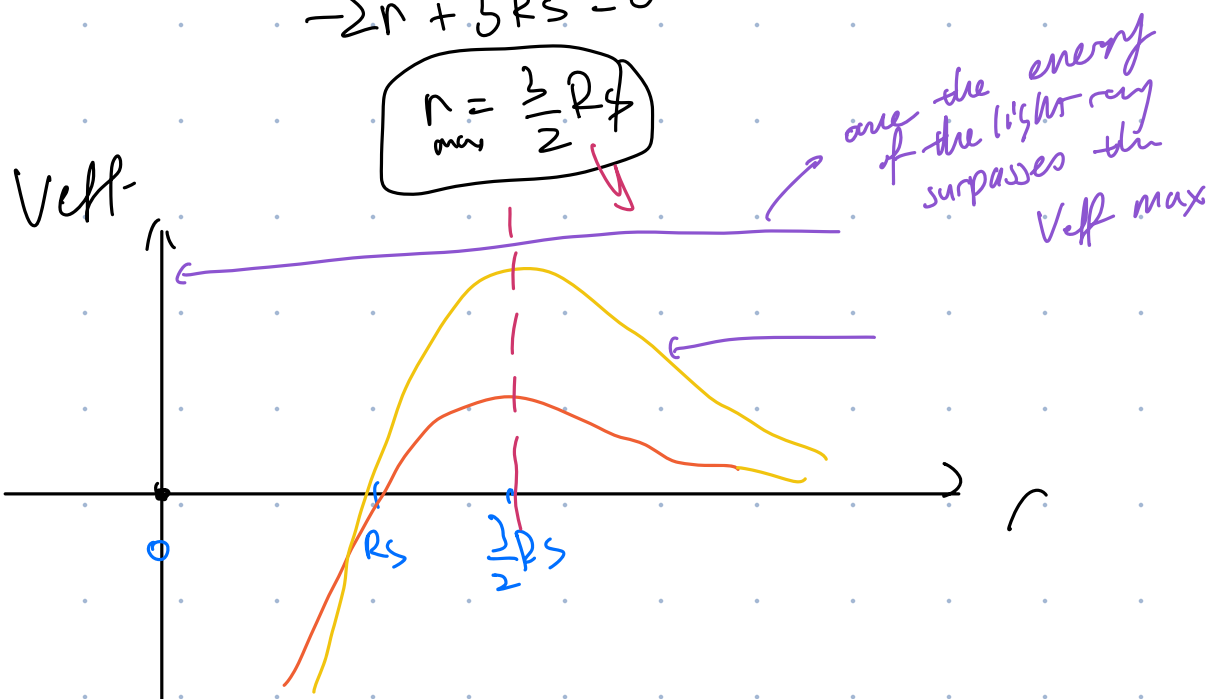
A Couple Examples:

$$V_{\text{eff}} = \frac{L^2}{r^2} \left(1 - \frac{R_S}{r}\right) \rightarrow \frac{1}{r^2} - \frac{R_S}{r^3}$$

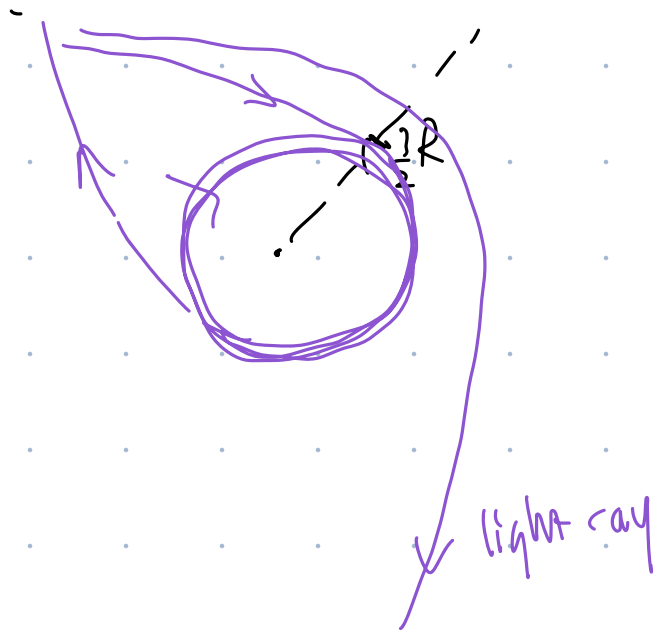
$$\frac{dV_{\text{eff}}}{dr} = -\frac{2}{r^3} + \frac{3R_S}{r^4} = 0$$

$$-2r + 3R_S = 0$$

$$r_{\text{max}} = \frac{3}{2} R_S$$



This is a mysterious
creature



Massive object:

$$b = 1 : v_{eff} = \left(\frac{L^2}{r^2} + 1 \right) \left(1 - \frac{R_s}{r} \right)$$

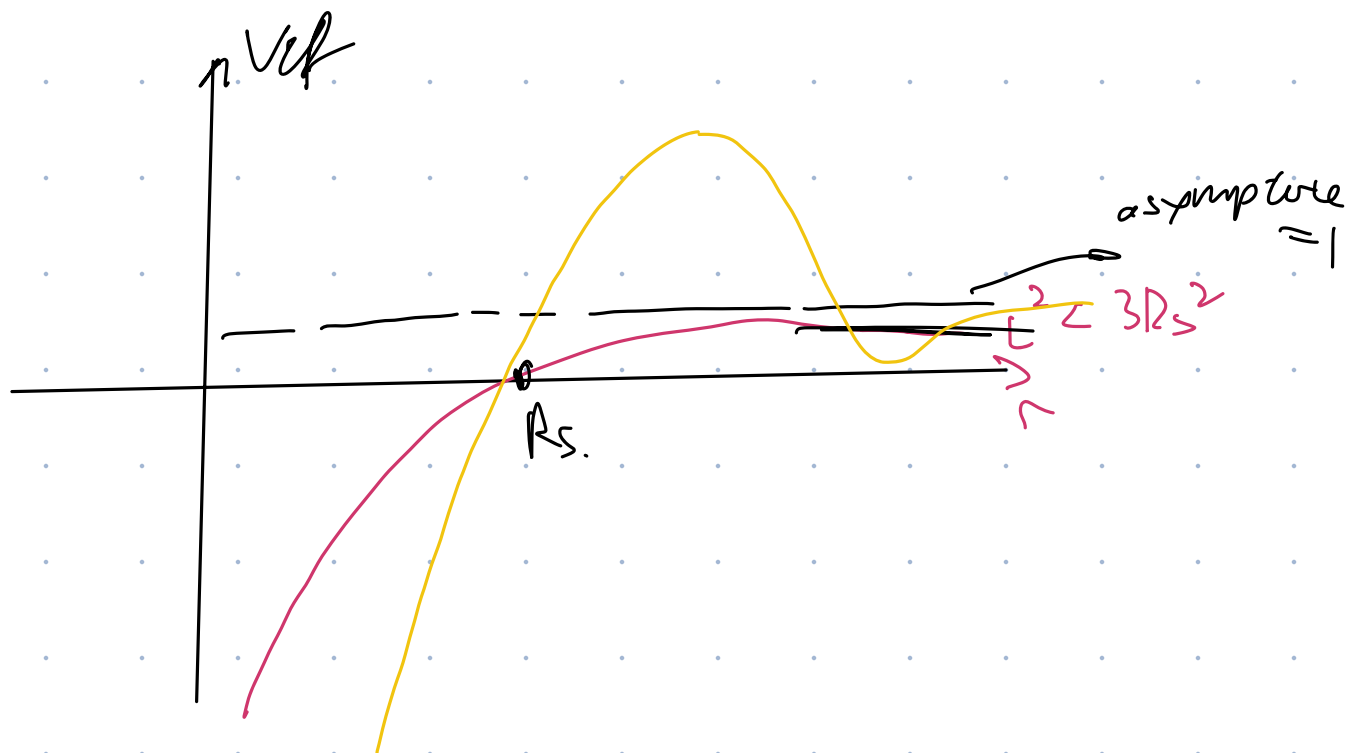
$$\frac{dv_{eff}}{dr} = -\frac{2L^2}{r^3} + \frac{3L^2 R_s}{r^4} + \frac{R_s}{r^2} = 0$$

could use
quadratic
formula:

$$R_s r^2 - 2L^2 r + 3L^2 R_s = 0$$

$$R_s \left(r - \frac{L^2}{R_s} \right)^2 = \frac{L^4}{R_s} - 3L^2 R_s$$

No min/max if $\frac{L^4}{R_s} < 3L^2 R_s$
OR $L^2 < 3R_s^2$



$$L = r^2 \frac{d\varphi}{dr}$$

$$\left(\frac{dr}{d\varphi} \right)^2 + V_{\text{eff}}(r) = E^2 \quad \left. \begin{array}{l} \text{trade } r(r) \text{ for } r(\varphi) \\ \frac{dr}{d\varphi} = \frac{dr}{d\varphi} \frac{d\varphi}{dr} = \frac{dr}{d\varphi} \frac{L}{r^2} \end{array} \right\}$$

$$\left(\frac{dr}{d\varphi} \right)^2 \frac{L^2}{r^4} + V_{\text{eff}}(r) = E^2$$

Standard Trick: $u \equiv \frac{2L^2}{R_s^2} \cdot \frac{1}{r}$

$$\frac{du}{d\varphi} = - \frac{2L^2}{R_s^2} \frac{1}{r^2} \frac{dr}{d\varphi}$$

$$\left(\frac{dr}{d\varphi} \right)^2 = \left(\frac{du}{d\varphi} \right)^2 \left(\frac{R_s^2 r^4}{4L^4} \right)$$

$$\left(\frac{du}{d\varphi}\right)^2 \left(\frac{R_s^2 r^4}{4L^4}\right) \left(\frac{L^2}{r^4}\right) + V_{eff}(u=\frac{1}{r}) = E^2.$$

$$\frac{1}{r} \rightarrow u. \quad \left[\left(\frac{du}{d\varphi}\right)^2 + u^2 - 2u - u^3 \frac{R_s^2}{2L^2} \right] = 4L^2(E^2 - 1)$$

Nonrelativistic

$$2 \frac{du}{d\varphi} \cdot \frac{d^2u}{d\varphi^2} + 2u \frac{du}{d\varphi} - 2 \frac{du}{d\varphi} - \frac{3}{2} \frac{R_s^2}{L^2} u^2 \frac{du}{d\varphi} = 0$$

$$\left[\frac{d^2u}{d\varphi^2} + u - 1 \right] - \left[\frac{3}{4} \frac{R_s^2}{L^2} u^2 \right] = 0$$

↑
harmonic
oscillator.

perturbation
term.

$$\left[\frac{d^2u}{d\varphi^2} = 1 - u \right]$$

↑
harmonic
oscillator

History of Measurement